From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FULL TEST – IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 04)** contains 4 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
   Section – A (05 – 09) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 mark** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.

3. **Section – C (01 – 08)** contains 8 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

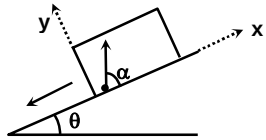
Physics

PART – I

SECTION – A
Single Correct Choice Type

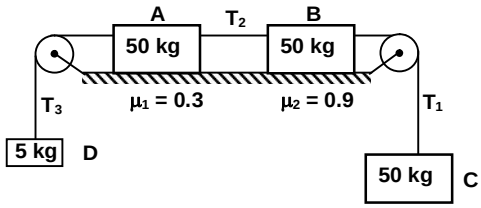
This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A large empty container is sliding without friction down a smooth inclined plane of inclination θ. From a point on the bottom of container a particle is projected with speed u with respect to box at an angle α with the bottom of container. The velocity $\vec{v}$ of the particle relative to the box after time t is



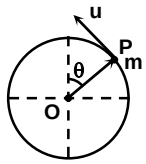
(A) $u\cos\alpha\,\hat{i} + (u\sin\alpha - gt)\,\hat{j}$
(B) $(u\cos\alpha - gt\sin\theta)\,\hat{i} + (u\sin\alpha - gt\cos\theta)\,\hat{j}$
(C) $u\cos\alpha\,\hat{i} + (u\sin\alpha - gt\cos\theta)\,\hat{j}$
(D) $u\cos\alpha\,\hat{i} + u\sin\alpha\,\hat{j}$

2. A block pulley arrangement is shown in the figure. If f_1 and f_2 represents frictional force acting on the block A and B respectively then choose the correct statement. (given g = 10 m/s^2)



(A) $T_1 = 100$ N, $T_2 = 10$ N
(B) $f_1 = 10$ N , $f_2 = 450$ N
(C) $T_2 = 50$ N, $T_3 = 50$ N
(D) $f_1 = 50$ N, $T_2 = 50$ N

3. A simple pendulum of bob of mass m and length ℓ has one of its ends fixed at the centre O of a vertical circle, as shown in the figure. If $\theta = 60°$ at the point P, the minimum speed u that should be given to the bob so that it completes vertical circle is



(A) $\sqrt{g\ell}$
(B) $\sqrt{2g\ell}$
(C) $\sqrt{\frac{5g\ell}{2}}$
(D) $\sqrt{\frac{3g\ell}{2}}$

Space for Rough work

4. A rod of mass m and length ℓ is lying along the y-axis such that one of its ends is at the origin. Suddenly an impulse is given to the rod such that immediately after the impulse, the end on the origin has a velocity $v_0\hat{i}$ and the other end has a velocity $2v_0\hat{i}$. The magnitude of angular momentum of the rod about the origin at this instant is

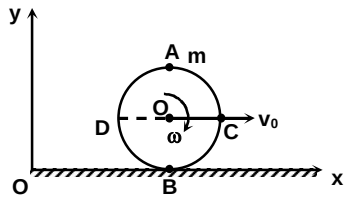
(A) $\frac{2}{3}mv_0\ell$ (B) $\frac{3}{2}mv_0\ell$

(C) $\frac{5}{6}mv_0\ell$ (D) $\frac{7}{8}mv_0\ell$

Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

5. A particle of mass 'm' is attached to the rim of a uniform disc of mass 'm' and radius R. The disc is rolling without slipping on a stationary horizontal surface, as shown in the figure. At a particular instant, the particle is at the top most position and centre of the disc has speed v_0 and its angular speed is ω. Choose the correct regarding the motion of the system (disc + particle) at that instant.



(A) $v_0 = \omega R$ (B) kinetic energy of the system is $\frac{11}{4}mv_0^2$

(C) speed of point mass m is less than $2v_0$ (D) $|\vec{v}_C - \vec{v}_B| = |\vec{v}_B - \vec{v}_D|$

6. Consider one dimensional collision between two identical particles A and B. B is stationary and A has momentum P before impact. During impact, B gives impulse $\vec{J}$ to A. Choose the correct alternative(s)

(A) the total momentum of the (A + B) system is P before and after the impact and (P – J) during the impact.

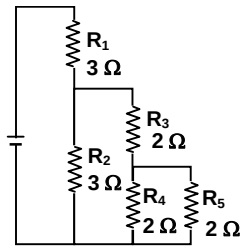
(B) During the impact, A give impulse $-\vec{J}$ to B.

(C) The coefficient of restitution is $\frac{2J}{P}-1$

(D) The coefficient of restitution is $\frac{J}{P}-1$

Space for Rough work

7. In the figure shown
(A) currents in R_4 and R_5 are equal
(B) current in R_2 and R_3 are equal
(C) current in R_1 is equal to the sum of currents in R_3, R_4 and R_5
(D) current in R_1 is equal to the sum of currents in R_2, R_4 and R_5



8. Energy liberated in the de–excitation of hydrogen atom from 3rd level to 1st level falls on a photo–cathode. Later when the same photo–cathode is exposed to a spectrum of some unknown hydrogen like gas, excited to 2nd energy level, it is found that the de–Broglie wavelength of the fastest photoelectrons, now ejected has decreased by a factor of 3. For this new gas, difference of energies of 2nd Lyman line and 1st Balmer line is found to be 3 times the ionization potential of the hydrogen atom. Select the correct statement(s) :
(A) The gas is lithium.
(B) The gas is helium.
(C) The work function of photo–cathode is 8.5 eV.
(D) The work function of photo–cathode is 5.5 eV.

9. The SI unit of inductance, 'henry' is the same as
(A) weber/ampere
(B) volt-second/ampere
(C) joule/(ampere)^2
(D) ohm-second

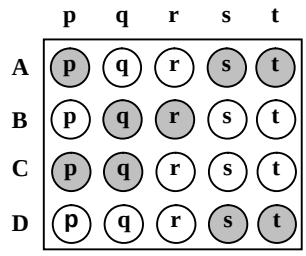
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SECTION - B

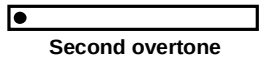
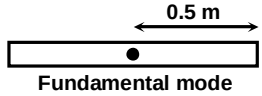
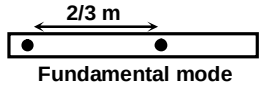
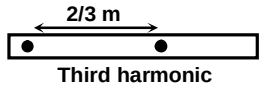
Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

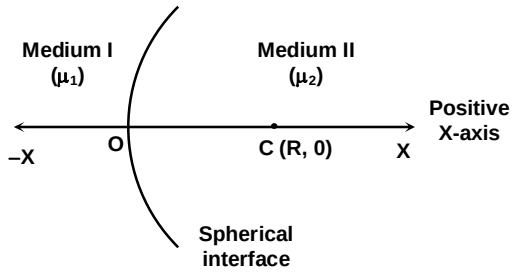


1. Longitudinal waves are produced in five identical rods in different situations. Their mode of vibration and fixed points (with bold point) are shown in the column II, and their frequencies are given in the column I, match the frequency in the column I with the appropriate situation in the column II. Each of the rods has length 1 m young's modulus of elasticity, 2×10^{12} N/m^2 and density 5×10^3 kg/m^3.

Column-I		Column-II	
(A)	10 KHz	(p)	 Second overtone
(B)	15 KHz	(q)	 0.5 m, Fundamental mode
(C)	25 KHz	(r)	 0.5 m, First harmonic
(D)	45 KHz	(s)	 2/3 m, Fundamental mode
		(t)	 2/3 m, Third harmonic

Space for Rough work

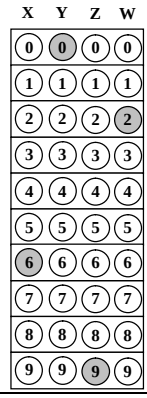
2. A spherical interface whose radius is R separates the two media, I and II as shown in the figure. The refractive index of medium I is μ_1 and that of medium II is μ_2. Now match the column I with column II [Consider the pole 'O' as the origin and principal axis as the x-axis]



Column I		Column II	
(A)	Object is kept at point (–a, 0) and $\mu_1 < \mu_2$	(p)	Image may be real
(B)	Object is kept at point (–a, 0) and $\mu_1 > \mu_2$	(q)	Image may be virtual
(C)	Object is kept at point (+a, 0) and $\mu_1 < \mu_2$	(r)	The position of image may be in between the position of object and origin.
(D)	Object is kept at point (+a, 0) and $\mu_1 > \mu_2$	(s)	The position of image may be in between the position of object and centre of curvature of interface.

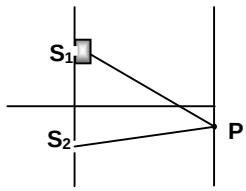
Space for Rough work

SECTION – C
Integer Answer Type



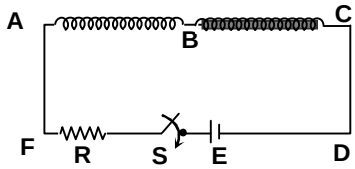
This section contains 8 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.

1. In a young's double slit experiment one of the slits is covered by a thin film of thickness t = 0.04 mm, and refractive index $\mu = 1.2 + \frac{9\times10^{-14}\text{m}^2}{\lambda^2}$, where, λ is the wave length in meter. A beam of light consisting two wavelengths λ_1 = 400 nm and λ_2 = 600 nm falls normally on the plane of the slits. Find the distance between two central maxima in milimeter. Distance of screen from slits is 400 times the separation between the slits.

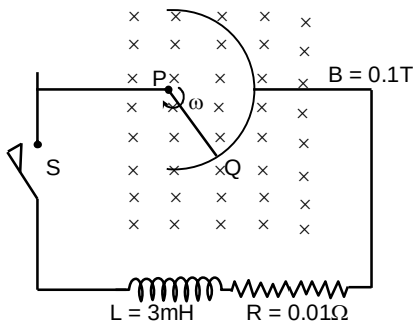


2. Two particles A and B are located at points (0, $-10\sqrt{3}$) and (0, 0) in xy plane. They start moving simultaneously at time t = 0 with constant velocities $\vec{v}_A = 5\hat{i}$ m/s and $\vec{v}_B = -5\sqrt{3}\hat{j}$ m/s, respectively. Time when they are closest to each other is found to be K/2 second. Find K. All distances are given in meter.

3. Two ideal solenoids of same dimensions. One is air cored with 600 turns while other is Aluminium cored with 200 turns (relative permeability of Aluminium is 3), are connected in a circuit as shown in the figure. The switch S is closed at t = 0. Find the ratio of potential difference across air-cored solenoid to that of Aluminium cored solenoid at any time t.



4. The circuit shows a resistance R = 0.01Ω and inductance L = 3mH connected to a conducting rod PQ of length ℓ = 2m which can slide on a perfectly conducting circular arc of radius ℓ with its centre at P. Assume that friction & gravity are absent and a constant uniform magnetic field B = 0.1T exists as shown in the figure. At t = 0, the circuit is switched on and simultaneously an external torque is applied on the rod so that it rotates about P with a constant angular velocity ω = 2 rad/sec. Find the magnitude of this torque (in N-m) at t = (0.3 ln2) second.

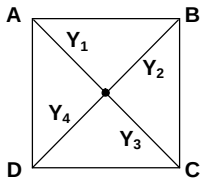


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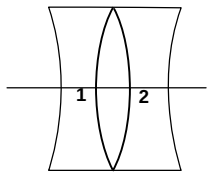
5. Two sound waves of frequencies 100 Hz and 102 Hz and having same amplitude 'A' are interfering. A stationary detector, which can detect waves of amplitude greater than or equal to A, So, in a given time interval of 12 seconds, find the total duration in which detector is active.

6. Two identical metallic sheets each behaves like a black body are kept parallel to each other with small separation between them in vacuum. Thermal energy is generated at a constant rate P in one of the sheets by a passing electricity in it. In steady state, temperature of the other sheet is found to be constant is 300K. Find the value of P in Kilo-Watt.
(Given area of the plates A = $\frac{10^3}{17\times 27}$ m^2 and $\sigma = \frac{17}{3}\times 10^{-8}$ w/m^{2}K^4)

7. In the adjacent figure, ABCD are four points in a smooth horizontal plane representing the four corners of a square of diagonal 20 cm. A very small object of mass 14 gm placed at the centre of the square connected to points A, B, C and D by four elastic strings of same dimensions but of different moduli of elasticity as shown in the figure. Natural length and cross-sectional area of each string are 10 cm and $\pi^2 \times 10^{-6}$ m^2 respectively. The mass is slightly displaced from equilibrium and then released to perform SHM along the diagonal AC. (Y_1 = 9Y, Y_2 = 4Y, Y_3 = 16Y and Y_4 = 4Y, where Y = 4.9 $\times$ 10^9 N/m^2). The time period of oscillation of object is k millisecond. The value of K is



8. Two identical equi-concave lenses made of glass of reflected index 1.5, placed in contact has power P. When a liquid of refractive index μ is filled in the gap between the concave lenses, the power becomes of initial value(P/3). The value of μ is $\frac{K}{3}$. Find the value of K.



Space for Rough work

Chemistry

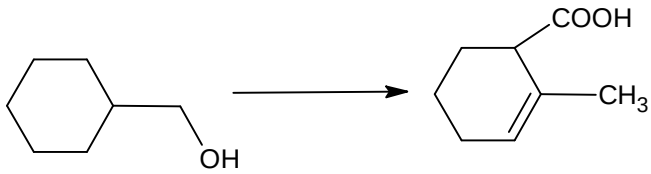
PART – II

SECTION – A

Straight Objective Type

This section contains **4 multiple** choice questions numbered **1 to 4.** Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. The correct order of reagents required to bring the following change is:



(A) H_3O^+ ; KCN ; NBS ; $K_2Cr_2O_7/H^+$ (B) $K_2Cr_2O_7/H^+$; Cl_2/KOH; CH_3MgBr ; H_3O^+

(C) conc. H_2SO_4 ; NBS ; KCN ; H_3O^+ (D) Conc. H_2SO_4 ; NBS ; CH_3MgBr ; $K_2Cr_2O_7/H^+$

2. 1 mole of a gas AB_3 present in 10 lt container at pressure 2.5 atm and 273 K temperature. On increasing the temperature to 546 K, AB_3 dissociates into $AB_2(g)$ and $B_2(g)$. If the degree of dissociation of AB_3 is 80%, then final pressure at 546 K is :

(A) 5 atm (B) 1.25 atm

(C) 10 atm (D) 6.25 atm

3. 1 mole of each of CaC_2, Al_4C_3, Mg_2C_3 reacts with excess water in separate open flasks work done during dissolution shows the order :

(A) $CaC_2 = Mg_2C_3 < Al_4C_3$ (B) $CaC_2 = Mg_2C_3 = Al_4C_3$

(C) $Mg_2C_3 < CaC_2 < Al_4C_3$ (D) $Mg_2C_3 < Al_4C_3 < CaC_2$

4. In which of the following pairs the two species are not isostructural?

(A) PCl_4^+ and $SiCl_4$ (B) PF_5 and BrF_5

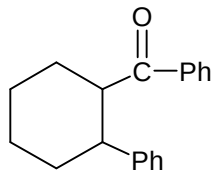
(C) AlF_6^{3-} and SF_6 (D) CO_3^{2-} and NO_3^-

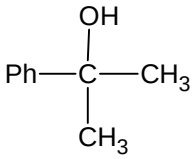
Rough Work

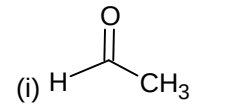
Multiple Correct Choice Type

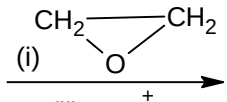
This section contains **5 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

5. Which of the following reactions are correctly matched with products (major)?

(A) PhMgBr $\xrightarrow[(ii)\ H_3O^+]{(i)}$ 

(B) PhMgBr $\xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3CN}$ (X) $\xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3Li}$ 

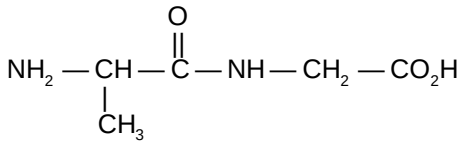
(C) PhMgBr $\xrightarrow[(ii)\ H_3O^+]{(i)}$ Pure enatiomer 

(D) PhMgBr $\xrightarrow[(ii)\ H_3O^+]{(i)}$ $PhCH_2CH_2OH$ 

6. For $I_2 + 2e^- \rightarrow 2I^-$, standard reduction potential = +0.54 volt.
For, $2Br^- \rightarrow Br_2 + 2e^-$, standard oxidation potential = – 1.09 volt
For, $Fe \rightarrow Fe^{2+} + 2e^-$, standard oxidation potential = +0.44 volt
Which of the following actions are spontaneous:
(A) $Br_2 + 2I^- \rightarrow 2Br^- + I_2$ (B) $Fe + Br_2 \rightarrow Fe^{2+} + 2Br^-$
(C) $Fe + I_2 \rightarrow Fe^{2+} + 2I^-$ (D) $I_2 + 2Br^- \rightarrow 2I^- + Br_2$

7. Which of the following will respond to positive chromyl chloride test?
(A) $CuCl_2$ (B) $ZnCl_2$
(C) $HgCl_2$ (D) AgCl

Rough Work

8. 

Identify the amino acids obtained by hydrolysis of the above compound :

(A) Glycine (B) Phenylalonine

(C) Alanine (D) Glutamic acid

9. Which of the following reaction produces N_2 ?

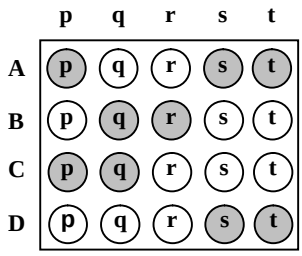
(A) $NH_4ClO_4 \xrightarrow[\Delta]{}$ (B) $KIO_3 + N_2H_4 \xrightarrow{H^+}$

(C) $HN_3 + HNO_2 \rightarrow$ (D) $NH_2OH + CuO \rightarrow$

Rough Work

SECTION-B
(Matrix Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the Column – I with Column – II:

Column – I(Ionic solid)	**Column – II(their Properties)**
(A) ZnS	(p) Limiting $\frac{r_+}{r_-} = 0.414$
(B) NaCl	(q) Frenkel Defect
(C) CsCl	(r) Crystallographic C.N = 4
(D) CaF_2	(s) Insoluble in water
	(t) 8 nearest neighbours of the cation

2. Match the Column – I with Column – II:

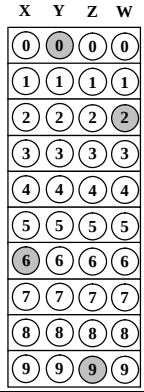
Column – I(compounds)	**Column – II(characteristics)**
(A) AsF_3	(p) sp^2
(B) $S_2O_3^{2-}$	(q) sp^3
(C) BF_3	(r) Back bonding
(D) $H_2C = SF_4$	(s) sp^3d
	(t) Coordinate bond

Rough Work

SECTION – C

Integer Answer Type

This section contains 8 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. The total number of cyclic, structural isomers for a compound with molecular formula C_5H_{10} is_____.

2. The equivalent weight of $Zn(OH)_2$ in the following reaction is equal to its M/x :

   $Zn(OH)_2 + HNO_3 \rightarrow Zn(OH)(NO_3) + H_2O$

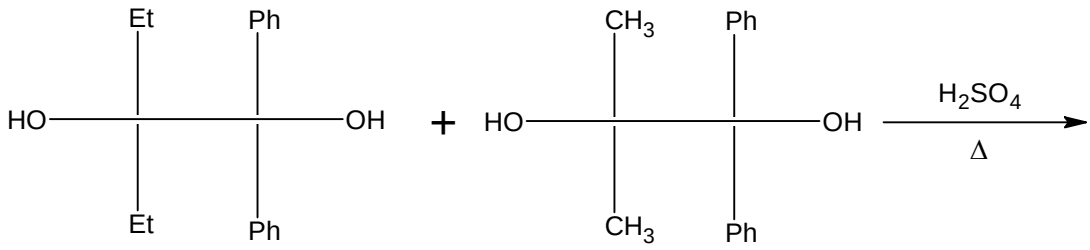
   X is equal to _____.

3. The number of ions formed when cuprammonium sulphate is dissolved in water is _____.

4. How many visible lines are emitted during transition from 5th orbit to ground state in hydrogen emission spectrum?

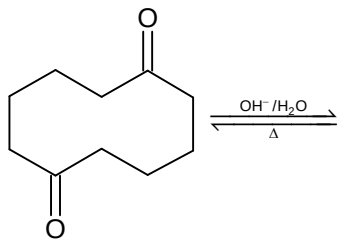
5. The half life of two samples of a substance is 50 sec and 200 sec at 0.4 M and 0.1M concentrations, respectively the order of reaction is _____.

Rough Work

6. 

How many products are obtained in the above reaction.

7. The total number of pi-bonds in pyrene responsible for its aromatic behaviour are:

8. 

Double bond equivalent of final product is .....

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **4 multiple choice questions** numbered 1 to 4. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If a_1, a_2, a_3 ..... are in A.P. and $b_k = a_k + a_{k+1} + ... + a_{k+n-1}$ (k = 1, 2, 3 .....), then $b_1 + b_2 + ... + b_n$ is equal to
   (A) $n(n + 1)a_n$ (B) $(n - 1)na_n$
   (C) n^2a_n (D) $(n + 1)^2a_n$

2. If for $|z - 1| = 1$, $z - 2 = kz \tan(\arg.z)$, then k is equal to
   (A) -1 (B) i
   (C) 1 (D) 2

3. Tangents PA and PB are drawn to the circle $x^2 + y^2 = 4$ from an external point P lying on the line y = x such that the angle θ between the tangents satisfies $\frac{\pi}{3} \le \theta \le \frac{2\pi}{3}$. The co-ordinates of the point P, if the perimeter of ΔPAB is maximum for all θ are
   (A) $\left(2\sqrt{2},\ 2\sqrt{2}\right)$ (B) $\left(\frac{2\sqrt{2}}{\sqrt{3}},\ \frac{2\sqrt{2}}{\sqrt{3}}\right)$
   (C) (2, 2) (D) none of these

4. The reflection of the complex number $\frac{2+i}{3-i}$ in the straight line $z(1 + i) = \bar{z}(1 - i)$ is
   (A) $\frac{2+i}{3+i}$ (B) $\frac{-2+i}{-3+i}$
   (C) $\frac{2-i}{3-i}$ (D) $\frac{2+i}{-3+i}$

Space for rough work

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

5. Two players A and B toss a coin which has a probability p of showing head ($0 < p < 1$), in the cyclic order A, A, B, A, A, B, ….. till a head appears. Let x denotes the probability that A gets the head first and y that of B, then
   (A) $x + y = 1$ (B) $x > y$ for all p
   (C) $x > \frac{1}{2}$ for all p (D) $y > \frac{1}{2}$ for all p

6. Let a, b, c be the sides of a triangle whose perimeter is P and area is A, then
   (A) $P^3 \le 27(b + c - a)(c + a - b)(a + b - c)$ (B) $P^2 \le 3(a^2 + b^2 + c^2)$
   (C) $a^2 + b^2 + c^2 \ge 4\sqrt{3}\,A$ (D) $P^4 < 256A^2$

7. A plane meets the co-ordinates axes in A, B, C such that the centroid of the triangle ABC is the point $(1, r, r^2)$. If all such planes must pass through the point (t, t, t) for all $r \in R - \{0\}$, then t can satisfy
   (A) $0 \le t \le 4$ (B) $1 < t \le 4$
   (C) $2 < t < 3$ (D) $0 < t \le 4, t \ne 3$

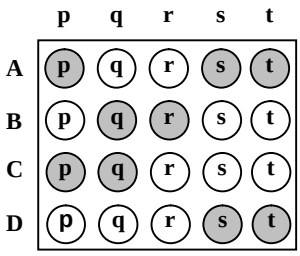
8. The value(s) of θ satisfying the equation $[\sin(\cot^{-1} \cos \tan^{-1} x)] = \sin \theta$, (where $x \in R$ and [.] denotes the greatest integer function) is/are
   (A) 0 (B) $\frac{\pi}{2}$
   (C) π (D) $\frac{3\pi}{2}$

9. Let $f(x) = (ax + b)\cos x + (cx + d)\sin x$ and $f'(x) = x \cos x$ for all $x \in R$, then
   (A) $a = -1$ (B) $b = 1$
   (C) $c = 1$ (D) $d = -1$

Space for rough work

SECTION-B
(Matrix Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	Two sides of a triangle are given by the roots of the equation $x^2 - 5x + 3 = 0$. If the angle between these sides is $\frac{\pi}{3}$, then the product of inradius and circumradius of the triangle is	(p)	12
(B)	In a triangle ABC, if the radii of ex-circles r_1, r_2 and r_3 are given by $r_1 = 8$, $r_2 = 12$ and $r_3 = 24$, then value of the side a is	(q)	15
(C)	Let a be a positive integer not exceeding 10, then the probability that the equation $x^2 - 2(a^2 - 1)x + 2a^2 - 7a + 3 = 0$ has one positive and one negative roots is	(r)	$\frac{3}{7}$
(D)	Let (α, β) be a point from which two perpendicular tangents can be drawn to the ellipse $4x^2 + 5y^2 = 20$. If $F = 4\alpha + 3\beta$, then the maximum value of F	(s)	$\frac{1}{5}$
		(t)	$\frac{2}{3}$

2. Match the following Column–I with Column–II

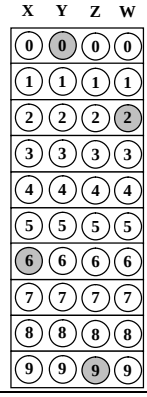
	Column – I		Column – II
(A)	The coefficient of two consecutive terms in the expansion of $(1 + x)^n$ will be equal, then n can be	(p)	9
(B)	If $15^n + 23^n$ is divided by 19, then n can be	(q)	10
(C)	${}^{10}C_0 {}^{20}C_{10} - {}^{10}C_1 {}^{18}C_{10} + {}^{10}C_2 {}^{16}C_{10} - \ldots..$ is divisible by 2^n, then n can be	(r)	11
(D)	If the coefficients of T_r, T_{r+1}, T_{r+2} terms of $(1 + x)^{14}$ are in A.P., then r is less than	(s)	12
		(t)	14

Space for rough work

SECTION – C

Integer Answer Type

This section contains **8 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ has a local maximum at $x = x_1$ and a local minimum at $x = x_2$ such that $x_2 = x_1^2$, then a is equal to _____

2. If positive numbers x, y, z are in A.P., then the minimum value of $\frac{x+y}{2y-x} + \frac{y+z}{2y-z}$ is equal to _____

3. The number of odd proper divisors of 24300 is N + 11, then N is equal to _____

4. The number of solutions of the simultaneous complex equations $|z - 3 - i| = 2$ and $|z - 2 + i| + |z - 4 - 3i| = 6$ is _____

5. The number of normal(s) to hyperbola which can touch its conjugate is equal to _____

6. Let z be a complex number satisfying $|z - 3| \le |z - 1|$, $|z - 3| \le |z - 5|$, $|z - i| \le |z + i|$ and $|z - i| \le |z - 5i|$. The area of the region in which z can lie is equal to _____

7. For $y = \cot^{-1}\left(\frac{\log ex^2}{\log e/x^2}\right) + \tan^{-1}\left(\frac{3+2\ln x}{1-6\ln x}\right)$, the value of $\frac{d^2y}{dx^2}$ at x = 1, is equal to _____

8. For real valued function f(x), $f(x) + f(x + 4) = f(x + 2) + f(x + 6)$, $g(x) = \int_x^{x+8} f(x)\,dx$. Then $g'(x)$ is equal to _____

Space for rough work